

GENERAL SIR JOHN KOTELAWALA DEFENCE UNIVERSITY

Faculty of Computing | Department of Computer Science

ADVANCED MOBILE COMPUTING

GROUP MINI-PROJECT PROPOSAL

CampusLink

Direct vs Epidemic Message Delivery in a Campus Walkway

Option F | Python-based mobility and message-forwarding simulation

SUBMITTED BY

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Group: CampusLink Team | Intake 41

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1. Project overview

CampusLink proposes a small Python simulation of short-range message exchange between people carrying mobile devices. The model places people (nodes) on a narrow campus-walkway grid. As nodes move, they can exchange stored messages when they meet. The purpose is to compare two forwarding strategies: direct delivery and epidemic forwarding.

In direct delivery, a message can pass from its original sender only to its intended receiver. In epidemic forwarding, a node may give a copy to another encountered node that does not already hold the message. The second approach may increase the chance of delivery, but it may also create many extra copies.

This is Option F in the assignment brief. It is a computer simulation, not an Android application or an experiment involving real phones. It is intended to provide repeatable evidence that both group members can explain during the live demonstration.

2. Research question and aim

Research question: How do direct and epidemic forwarding compare in message delivery percentage, delivery time and number of copies as the number of mobile nodes changes in a campus-walkway simulation?

Aim: To develop a simple, repeatable simulator that demonstrates how node density and forwarding strategy influence opportunistic message delivery.

3. Project objectives

- Build a virtual walkway in which mobile nodes move and detect nearby nodes.
- Implement direct delivery and epidemic forwarding using simple, explainable rules.
- Compare delivery percentage, delivery time and message-copy count for three group sizes.
- Automatically save data and present a summary table and at least one comparison chart.

4. Expected findings and scope

We expect epidemic forwarding to deliver a higher percentage of messages, often in fewer simulation steps, while creating more copies than direct delivery. Larger groups may produce more encounters, although the measured results may differ from these expectations.

The model studies store-carry-forward messaging and encounters in a simplified virtual campus walkway. It will not perform actual Wi-Fi or Bluetooth communication, collect student-location data or predict the performance of a real campus network.

5. Proposed system and operation

The simulator will use Python 3 and run on a computer. It will place 10, 20 or 40 nodes on an 18-column by 6-row grid. Six messages, each with a source and a different destination, will be created for each run. Each run will simulate 180 discrete time steps.

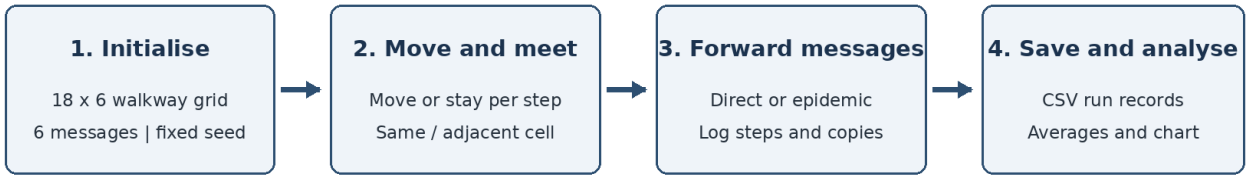


Figure 1. Proposed CampusLink workflow from seeded setup to exported measurements.

5.1 Procedure for one run

- Initialise positions and six source-destination pairs using a fixed random seed.
- At each step, let each node stay or move one cell horizontally or vertically within grid boundaries.
- Treat nodes as meeting when they occupy the same cell or horizontally/vertically adjacent cells.
- For direct delivery, transfer a message from its original source only to its destination. For epidemic forwarding, allow a holder to copy it to a new holder during a meeting.
- Record each first successful delivery and every new transfer. Once a message arrives, stop forwarding that message. Continue the run through the configured 180 steps and save results.

5.2 Software and technologies

Component	Planned use
Python 3 (standard library)	Movement, encounter detection, forwarding rules and command-line demonstration.
random.Random	Fixed random seeds for matching initial positions, messages and movement across strategies.
csv and datetime	Save run records, message outcomes and an experiment-session timestamp.
Microsoft Excel / compatible spreadsheet	Show group-wise averages and a delivery-percentage comparison chart.

6. Experiment and measurement plan

The independent settings are group size and forwarding method. The walkway size, movement rule, encounter rule, number of messages and time-step limit will be kept constant. For matched runs, both strategies will use the same seed so that their starting conditions and movements can be compared fairly.

Parameter	Planned setting
Virtual space	18 x 6 cells (narrow campus-walkway layout)
Situations	10, 20 and 40 mobile nodes
Routing strategies	Direct delivery and epidemic forwarding
Repetitions	10 runs per group size and strategy: 3 x 2 x 10 = 60 total runs
Per-run settings	6 messages; maximum of 180 simulated time steps
Encounter rule	Same grid cell or one horizontally/vertically adjacent cell
Reproducibility	Fixed, recorded seeds; matched seeds for both routing strategies

6.1 Measurements

Delivery percentage: (Number of messages delivered / 6) x 100. Delivery time: mean number of simulation steps among successfully delivered messages; if none arrive, the value is left unavailable. Copies created: count of successful transfers to new message holders (including delivery to the destination).

The simulator will automatically export 60 run-level records with settings, seeds, metrics and a recorded experiment-session timestamp, together with message-level logs. We will compare the six configurations in a spreadsheet and chart delivery percentage against group size for both methods.

7. Checkable deliverables

- Demonstrate direct and epidemic forwarding using 10, 20 and 40 nodes.
- Export 60 recorded simulation runs to CSV with seeds, timestamps and measured results.
- Present an average-results table and comparison chart, and rerun a selected experiment live.

8. Limitations and validation

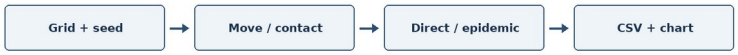
A random walk is not the same as real student movement; a short time limit and ten repeats per setting may affect conclusions. We will test movement boundaries, encounter detection, forwarding rules and repeatability. No names or identifiers of real devices or students will be collected.

9. Proposed work plan after approval

First, both members will check the movement, contact and routing functions with small test cases. Next, we will run and save the 60 experiments using the documented seeds, produce the spreadsheet table and chart, and review whether the findings match our expectation. Finally, we will collect two genuine screenshots from the running program, complete the required 2-3 page report and submission files, and practise independent live demonstrations.

Appendix A. Completed proposal form

The required fields from Appendix A of the assignment brief are completed below, together with the proposed workflow.

Appendix A item	Our proposed answer
Project title and option letter	CampusLink: Direct vs Epidemic Message Delivery in a Campus Walkway (Option F).
Group name, member names and registration numbers	CampusLink Team; PHMLCB Abeyrathne (D/BSE/24/0018); MAH Dharmasiri (D/BCS/24/0009).
Our question (one sentence)	How do direct and epidemic forwarding compare in delivery percentage, delivery time and copies for different group sizes?
What we expect to find (one sentence)	Epidemic forwarding may improve message delivery and delivery time, but create more copies than direct forwarding.
What our app or program will do (3 to 5 points)	1. Move nodes on a walkway grid. 2. Detect encounters. 3. Apply both forwarding rules. 4. Record message outcomes. 5. Export results to CSV.
Situations compared and repetitions	Three group sizes (10, 20, 40), both rules, ten repeats per setting; 60 runs in total.
Phone / emulator / simulator	Python 3 simulator on a computer; no phone or emulator is needed for Option F.
Our customisation	Narrow 18 x 6 campus-walkway grid with matched fixed-seed movement for fair comparison.
Three checkable deliverables	1. Demonstrate both strategies at all three group sizes. 2. Export 60 timestamped CSV run records. 3. Show a summary table and chart; rerun one selected experiment live.
Block diagram	

This detailed proposal and completed Appendix A are submitted together for lecturer review. Implementation and final measurements will follow the approved scope.